

### Intro (Level 0)

**Q1.** What is  $2^3$  equal to?

- (A) 6
- (B) 7
- (C) 8
- (D) 9
- (E) 10

**Q2.** What is  $x^2$  when  $x = 3$ ?

- (A) 6
- (B) 7
- (C) 8
- (D) 9
- (E) 10

**Q3.** What is the value of  $5^1$ ?

- (A) 1
- (B) 5
- (C) 10
- (D) 25
- (E) 50

**Q4.** What is  $3^2$  equal to?

- (A) 6
- (B) 7
- (C) 8
- (D) 9
- (E) 10

**Q5.** What is the meaning of  $x^0$  when  $x \neq 0$ ?

- (A) 0
- (B) 1
- (C)  $x$
- (D)  $-x$
- (E)  $\frac{1}{x}$

**Q6.** What is  $2^4$  equal to?

- (A) 8
- (B) 12
- (C) 16
- (D) 20
- (E) 24

**Q7.** What is  $1^5$  equal to?

- (A) 0
- (B) 1
- (C) 5
- (D) 10
- (E) 15

**Q8.** What does the term  $a^m$  represent?

- (A)  $a$  multiplied by itself  $m$  times
- (B)  $a$  divided by itself  $m$  times
- (C)  $a$  added to itself  $m$  times
- (D)  $a$  subtracted from itself  $m$  times
- (E)  $a$  raised to the power of  $m - 1$

#### Open Ended Questions (FRQ)

**Q9.** Explain what the exponent in  $x^2$  represents and provide an example.

**Q10.** Describe a scenario where you would use exponents in real life, such as in science or finance.

#### Chef's Tips

- Emphasize the concept that exponents represent repeated multiplication, as this is fundamental to understanding more complex exponent rules.
- Use visual aids such as graphs or blocks to help students visualize the concept of exponents and how they grow.
- Encourage students to explore real-world examples where exponents are used, such as population growth, chemical reactions, or financial calculations, to make the learning experience more engaging and relevant.